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A palaeomagnetic and geochemical study of basic intrusions in northern Sweden

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Abstract: A palaeomagnetic and geochemical study has been performed on basic dykes in northern Sweden. The dykes and a gabbro formation were sampled in 28 sites and characteristic magnetizations could be defined in 23 of them. The dykes form a part of a swarm that trends in NE–SW to E–W. From differences in palaeomagnetic signatures and composition it is concluded that this swarm is composed of two generations of dykes, group A and B, trending in similar directions. The dykes of group A have compositions that are similar to rapakivi related dykes, while those of group B are different from most rapakivi dykes in Fennoscandia. The calculated pole positions may suggest that the group B dykes are older than those of group A and both groups intruded within the time span 1.77 Ga to 1.50 Ga. The trend of the dykes is more or less parallell to a palaeo-compressional stress field that may be expected from the collisional tectonics related to the Gothian orogeny. The intrusion of the rapakivi formations in Fennoscandia has been suggested to be related with the Gothian orogeny and the intrusion of the dykes may thus be guided by the stress field generated by the collisional tectonics.

Keywords: Palaeomagnetism, plate tectonics, geochemistry, basic dykes, northern Sweden, Baltica.

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Introduction

Basic dykes are generated from crustal extension of both oceanic and continental environments and represent a channel by which basaltic magma is transferred from the mantle to the upper crust. Basic dykes have been intruded occasionally in continents during the Earth´s history and for many dyke swarms several percent of crustal extensions of a continent are the result of huge volume of basic magma that is added to the crust. The dykes often contain magnetite and titanomagnetite and the fast cooling make them good recorders of the palaeomagnetic field. They account for more then 30% of all palaeomagnetic poles, that have been obtained from Precambrian rocks (Halls & Fahrig 1987).

Aeromagnetic data and geological mapping in central Sweden show a system of dykes trending NE–SW to E–W (Figs. 1 and 2) and with this study we want to put these intrusions into a regional geological and plate tectonic context by making a palaeomag-netic and geochemical investigation. We also want to test the idea of these dykes being related to rapakivi intrusions in Sweden as suggested by Gorbatschev et al. (1987).

Geology

The Fennoscandian Shield can be divided into different tectonomagmatic blocks (Fig. 1). These blocks reveal a pattern of geochronological zonation with decreasing ages of the rocks from the northeast to the southwest. This zonation reflects a soutwestward growth of the Fennoscandian Shield during successive orogenic events in the Precambrian, i.e. the Lopian (2.9–2.6 Ga), the Svecofennian (1.9–1.85 Ga), the Gothian (1.75–1.55 Ga) and the Sveconorwegian (1.25–0.8 Ga) events. The Sveco-fennian province is divided into northern, central and southern subprovinces (Gaal & Gorbatschev 1987; Fig. 1).

In the southern subprovince the basement rocks, in a large area covered by platform sediments, characterize a gradual growth of continental crust between 2.0 and 1.85 Ga (Bogdanova et al. 2001). This gradual growth is the result of subduction-collisional accretion of juvenile crust to the Fennoscandia and Sarmatia. The final accretion of the Ukrainian Shield and Sarmatia to Fennoscandia took place 1.8 to 1.7 Ga ago (Elming et al. 2001).

Precambrian mafic dyke swarms of the Fennoscandian Shield have been classified into 4 groups (Gorbatschev et al. 1987). These groups refer to: (1) rifting during the opening of the pre-Caledonian Iapetus Ocean and the intrusion of dykes in Late Precambrian, (2) those associated with hypabyssal sheet–sill complexes formed during various stages of the Sveconorwegian-Grenvillian orogeny, (3) the intrusion of dyke swarms at 1.7–1.5 Ga ago during the formation of new continental crust in western Scandinavia and the intrusion of rapakivi granites, and (4) the rifting of the Archaean shield and the intrusion of dykes between 2.40 and 1.85 Ga. Gorbatschev et al. (1987) suggested that the consolidation of a large segment of the Fennoscandian Shield

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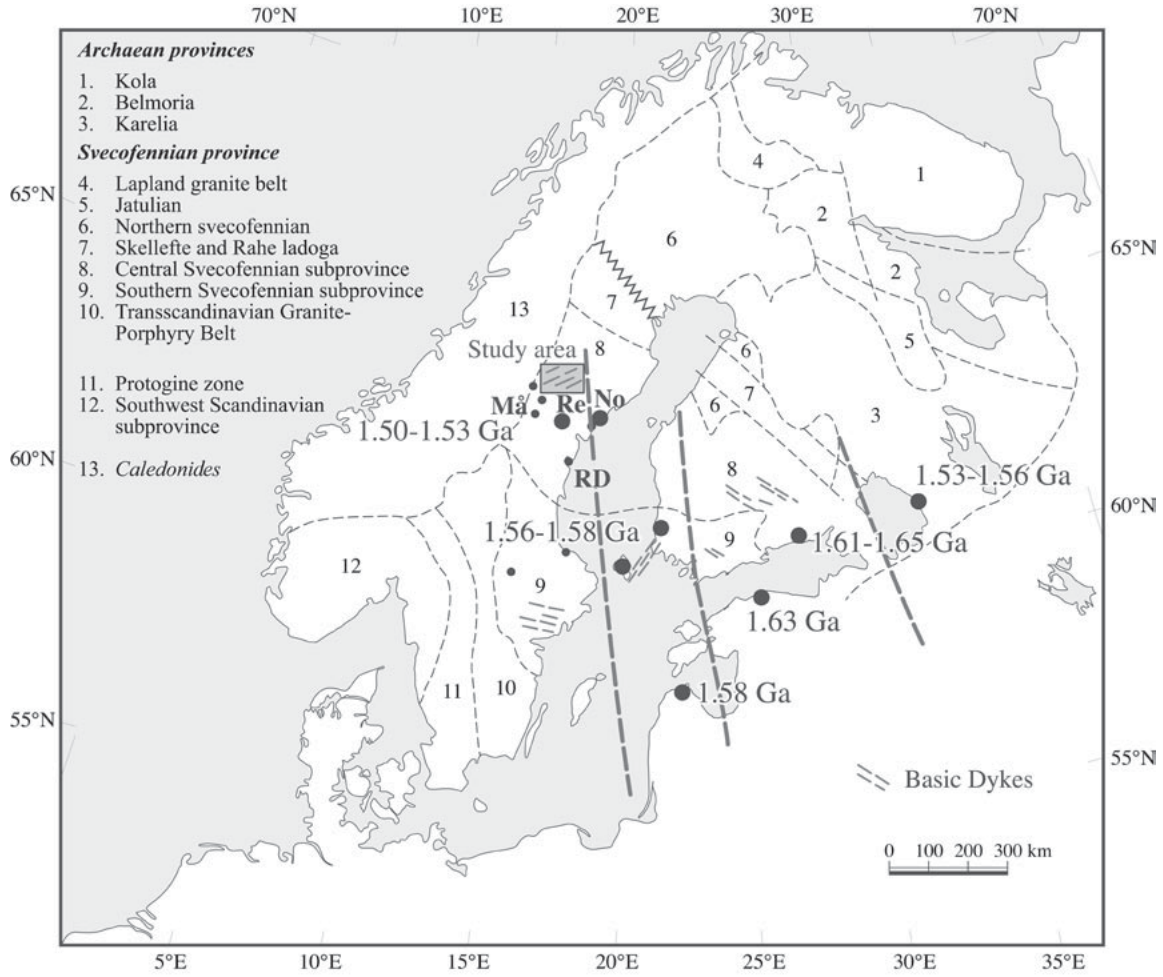


Fig. 1. The Fennoscandian Shield and the tectonomagmatic block division (modified after Gaal & Gorbatschev 1987; Pesonen et al. 1989). The rapakivi intrusions are marked with dots and the minor intrusions are marked with smaller symbols. Intrusions in the vicinity of the study area (defined by the frame) are the Nordingrå (NO), the Ragunda (Re) and the Mårdsjö (Må) formations. Further south is the Rödö (RD) formation located. The grey hatched lines mark zones of different generations of rapakivi intrusions.

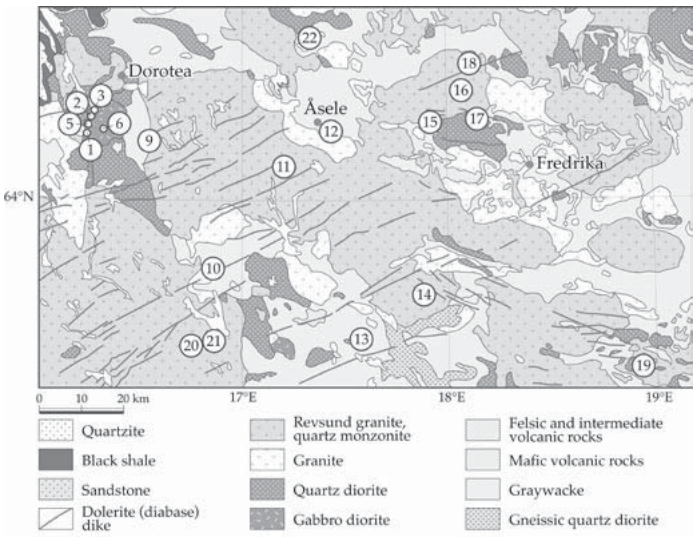


Fig. 2. Geological map of the study area (modified after Lundqvist et al. 1990). The numbers denote the sites where characteristic magnetizations have been isolated in the dykes and in the gabbro.

during 1.85 Ga to 1.25 Ga stimulated fracturing and provided a good condition for dyke intrusion.

The region of this study is a part of the central Svecofennian subprovince (block 8 in Fig. 1), where metasediments and Revsund granite are the dominating rocks. A gabbro formation is located in the western part of the study area. This gabbro is believed to belong to the Svecofennian orogeny (c. 1.87 Ga; Lundqvist et al. 1990), which is characterized by very large volumes of igneous rocks produced within a short time and by a predominance of granodiorites and tonalites (Gaal & Gorbatschev 1987). There are a number of c. NE–SW trending basic dykes in the area (Fig. 2) and some of these dykes cut the gabbro. The width of these dykes varies between c. 1.0 m and 1.5 m.

In the south and southeast of the area small rapakivi complexes have intruded into the 1.8 to 1.77 Ga old Revsund granite. These intrusions form a part of a discontinous belt of anorogenic granite plutons running from western Russia over southern Finland, Estonia and Latvia to central Sweden. Apart from the Salmi batholite in Russia, there is a westward trend of decreasing age of the complexes from the Wiborg batholite (1.65–1.62 Ga; Vaasjoki et

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al. 1991) in the east to the small complexes in central-northern Sweden (1.53–1.50 Ga; Andersson 1997a) in the west.

Sampling and laboratory technique

The palaeomagnetic samples were collected with a portable drilling machine and as block samples and oriented with sun and magnetic compasses. Totally 136 samples were collected at 22 sites from the basic dykes and at 6 sites from the gabbro. For the chemical analyses specimens from the drillcores and block samples were used.

The intensity and direction of magnetization was measured with a spinner (JR5, AGICO) and a SQUID (2G-DC) magnetometer. The samples were demagnetized in alternating field in steps of 5 mT up to 40 mT and 10 mT steps up to 100 mT. Thermal demagnetizations were made in temperatures up to 620°C with intervals of 50°C and 10–20°C for low (up to 500°C) and high temperatures, respectively (TSD1, Shonstedt). Mean susceptibility, anisotropy of susceptibility and susceptibility–temperature variations were measured with a kappabridge (KLY-3 and CS-3, AGICO). For the latter measurements, pieces of rocks were measured during warming and cooling from 25 to 700°C.

Components of remanent magnetization were isolated using a methods of least square fit of vectors in three dimensional space (Torsvik 1986; Torsvik et al. 1996) and by principal component analyses (Kirschvink 1980). Remagnetization circles (Halls 1976) have also been used to define the individual directions. The majority of remanence directions have been defined with a maximum angular deviation (MAD) lower than 7°.

Palaeomagnetic and rock magnetic results

Magnetic susceptibility and Q-values

The susceptibility of the basic dykes varies in a wide range between $0.49 \cdot 10^{-3}$ and $84.6 \cdot 10^{-3}$, with an average of $30.4 \cdot 10^{-3}$. The maximum degree of anisotropy (P), foliation (F) and lineation (L) is 19%, 5% and 5%, respectively. The anisotropy indicates that the dykes are either magnetically foliated, lineated or neutral. In most cases the minimum axis of suceptibility is almost horizontal. The plane of maximum and intermediate susceptibility axes, the foliation plane, is perpendicular to the minimum axis. The plane of foliation is related to the flow of the magma and therefore expected to be parallel to the dyke plane (Tarling & Hrouda 1993), which means that the dykes here are generally vertical.

On basis of susceptibility and Q-values the dykes can be divided into four groups: (1) sites 11 and 17, (2) sites 2, 9, 10, 12, 13, 15, 16 and 22, (3) sites 14, 18, 20, and 21, (4) site 3. Dykes of group 1 have susceptibilities and NRM values that are lower than those of group 2, while the susceptibility and Q-values of group 3 are similar to and overlapping with those of the groups 1 and 2. The dyke at site 3 also has susceptibilities overlapping those of group 1 and 2. The Q-values are generally between 0.1 and 3.0, however, the dyke at site 3 has Q-values as high as 40.0.

The susceptibility of the gabbro samples varies between $0.486 \cdot 10^{-3}$ and $42.3 \cdot 10^{-3}$ with an average of $12.09 \cdot 10^{-3}$. The average degree of anisotropy (P) is c. 20% and the rock is predominantly magnetically foliated. The mean direction of the minimum axes of susceptibility has a declination of 250° and an inclination of 30°. The directions of the maximum and intermediate axes are dispersed and the mean directions are not well defined.

The magnetic foliation has probably only little influence on the direction of the remanent magnetization as the maximum deflection of the palaeofield for 20% of anisotropy is expected to be not more than c. 5° (McElhinny 1973).

Palaeomagnetic results

Characteristic magnetizations in the basic dykes have been identified in seventeen of the twenty sites. Generally, this magnetization was isolated in AF stronger than 20 mT and at temperatures higher than 500°C (Fig. 3). Only in site 15 a stable magnetization is defined from magnetizations of lower coercivities (AF≥15 mT) and in the sites 16 and 21 the characteristic magnetizations are isolated in temperatures of 400°C and higher.

Two main groups of site mean directions have been defined (A: sites 2, 3, 20a, 21; B: sites 6, 9, 10, 11, 12, 14, 16, 17, 18, 22; Fig. 4B; Table 1) and a separate group of widely dispersed directions (sites 13, 15, 19, 20b). In three of the dykes (sites 4, 7, 8) no stable remanent magnetization could be defined. There is no strict relation between the three groups of remanence directions and the difference in NRM, Q-value and susceptibility, which means that dykes within the same group are not magnetically homogeneous.

The trend of the dykes of group A is NE to ESE and the mean direction of the characteristic magnetization is decl. = 44.8° and

Table 1. Site mean directions and calculated pole positions from the basic dykes in this study.

Site	Lat./Lon.	N	Decl.	Incl.	α_{95}	K	Plat.	Plon.	A_{95}	Gr
Basic dykes										
2	64.2/16.3	8	53.8	–9.8	12.1	22	10.3	141.5	8.7	A
3	64.2/16.3	20	52.6	19.0	8.4	16	24.4	137.0	6.4	A
6	64.1/16.3	3	2.2	19.6	15.7	63	36.0	193.6	11.9	B
9	64.1/16.6	9	356.8	58.3	8.5	38	64.8	202.4	10.8	B
10	63.8/16.9	6	1.6	34.9	16.5	17	45.4	194.7	14.4	B
11	64.1/17.3	7	1.1	35.1	10.0	38	45.3	195.8	8.7	B
12	64.2/17.4	5	10.7	54.2	10.7	52	59.8	179.7	12.6	B
13	63.7/17.5	7	334.7	29.8	12.5	24	39.2	229.5	10.3	
14	63.8/17.9	5	4.4	42.0	8.7	78	50.3	191.6	8.3	B
15	64.2/17.9	6	82.1	43.5	21.9	10	26.1	103.1	21.5	
16	64.2/18.0	9	1.0	20.5	8.7	36	36.4	196.8	6.6	B
17	64.2/18.2	4	1.0	28.0	10.3	81	40.7	196.9	8.4	B
18	64.3/18.1	7	13.1	21.9	6.9	66	36.3	182.1	5.3	B
19	63.7/18.8	4	193.8	26.3	20.9	20	11.7	185.1	16.2	
20a	63.7/16.8	3	41.4	12.2	10.0	154	25.3	150.1	7.3	A
20b	63.7/16.8	3	8.1	2.7	15.2	67	27.4	187.7	10.7	
21	63.7/16.8	6	31.4	11.2	9.6	49	27.7	160.9	6.9	A
22	64.3/17.3	5	11.3	36.6	9.9	61	45.4	182.1	8.8	B
Mean Group A		**4*/37**	**44.8**	**8.3**	**18.7**	**25**	**22.1**	**147.2**	**13.4**	
Mean Group B		**10*/60**	**4.4**	**35.2**	**8.5**	**33**	**45.1**	**191.6**	**7.5**	
Gabbro										
1	64.1/16.2	7	315.5	39.4	7.5	65	39.0	252.8	7.0	
2	64.2/16.3	9	336.6	33.2	9.1	33	41.3	226.4	7.8	
3	64.2/16.3	16	334.5	42.5	6.5	50	47.1	231.4	6.3	
5	64.1/16.2	7	319.3	42.3	10.9	32	42.4	249.7	10.5	
6	64.1/16.3	10	341.3	29.1	7.1	62	39.8	220.0	5.8	
Mean		**5*/49**	**329.9**	**37.7**	**10.3**	**57**	**42.6**	**235.7**	**9.3**	

Glat/Glon – geographical latitude and longitude of the site; N – number of samples; Decl. – mean declination; incl. – mean inclination; * indicates if the mean has been calculated from site means or from samples; a_9 – radius of 95% confidence circle; Plat., Plon. – latitude and longitude of calculated palaeomagnetic pole position; K – Fisher (1953) precision parameter; A_{95} – radius of the circle of 95% confidence of the pole; Gr – the groups to which the dykes are refered.

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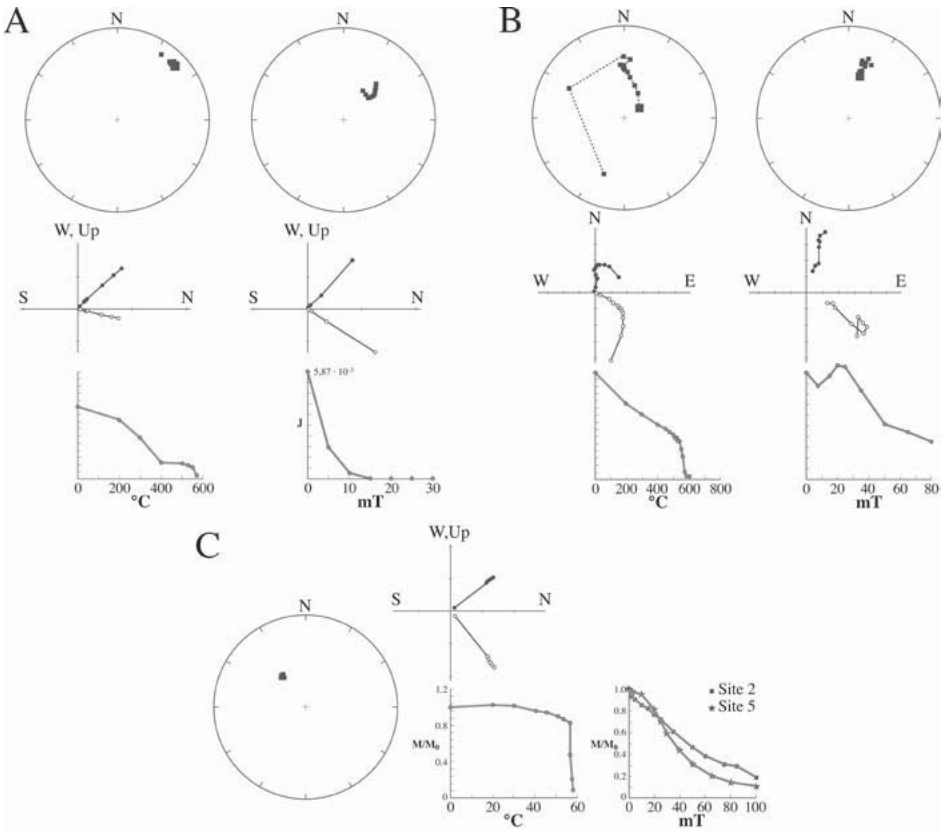


Fig. 3. Exemples of remanence behaviour during alternating field and thermal demagnetizations of samples from: **A.** the group A dykes, **B.** the group B dykes and **C.** the gabbro. The remanence directions are shown in stereographic projections and in vector plots. Note the stability of directions at high alternating fields and temperatures.

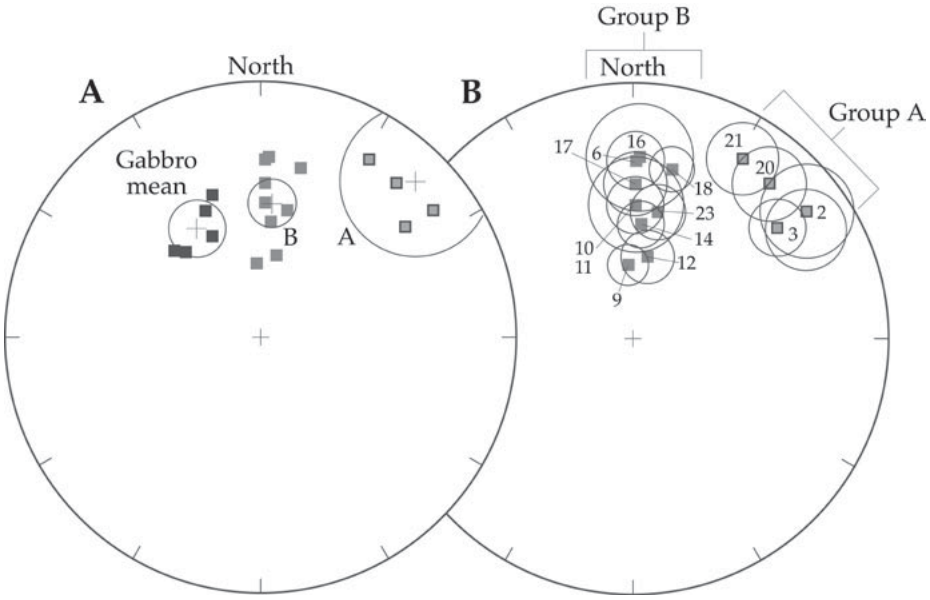


Fig. 4. **A.** Formation means of group A and B dykes and the gabbro. **B.** Site mean directions of characteristic magnetizations and associated α_{95} confidence circles for the basic dykes.

incl. = 8.3° (Fig. 4A; Table 1). At least for one of the dykes (site 3) a positive contact test can be demonstrated (Elming & Moakhar, in prep.). In the basic dykes at sites 2, 20 and 21 of group A the main magnetic carrier is magnetite-titanomag-netite (Fig. 5). In site 3 of the same group pyrrhotite and magnetite are the dominating magnetic minerals, as indicated by the temperature-susceptibility variations and thermal demagnetizations (Figs. 3A and 5).

The palaeomagnetic pole position (Fig. 6B) calculated from the poorly defined mean direction (decl. = 44.8°, incl. = 8.3°, α_{95} = 18.7°; Table 1) of group A is: Plat. = 22.1°, Plon. = 147.2°, (A_{95} = 13.4°).

The dykes of the group B also trend from NE to ESE and most of them are vertical. This is also indicated by the nearly horizontal minimum axes of susceptibility, except for the dyke at site 17 where the intermediate principle axis is horizontal and perpendicular to the dyke plane. Magnetite-titanomagnetite is the main magnetic carrier in the dykes of group B. The range of the unblocking temperatures is wide, however, 40–50% of the magnetization is still left at c. 570°C and the characteristic magnetization is often defined in a narrow temperature range (560–580°C; Fig. 3B). In the low coercivity and unblocking temperature ranges most of the dykes carries a magnetization with a direction similar to that of the PEF (decl. = 0°, incl. = 75°). The high unblocking temperature components have northerly declinations and intermediate to high positive inclinations. The site mean declinations vary between 357 and 13°, while the inclinations vary between 10 and 58° and some mean directions are significantly different (Fig. 4B). A possible reason for the variation in inclination may be due to the shape of the dykes, i.e. the demagnetization effect of a thin vertical sheet may cause the remanence direction to deviate towards the plane of the dyke. However, the susceptibilities of the dykes are not extreemly high (c. $30 \cdot 10^{-3}$) and the lack of correlation between the directions of magnetization and trends of the dykes indicates that they are not influenced by a demagnetizing effect of the dykes (Bates & Halls 1990). Another possible explanation for these variations in inclination is secular variations in the Earth magnetic field.

The mean direction of the group B dykes (decl. = 4.4°, incl. = 35.2°; Table 1) is fairly well defined (α_{95} = 8.5°) and significantly different from that of group A (Fig. 4A). The pole position is calcu-

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lated at Plat. = 45.1° and Plon. = 191.6° (A_{95} = 7.5°).

There are four site mean directions that are different from most of the other site means (Table 1). Magnetite is the main carrier of remanence in these dykes and the unblocking temperature ranges are generally wide. One of the site means (site 13: decl. = 334.7°, incl. = 29.8°, α_{95} = 12.5∞; Table 1) is very similar to the formation mean direction of the gabbro (decl. = 332.9°, incl. = 37.7°, α_{95} = 10.3°; Fig. 4A) and the dyke is probaly intruded at the same time as the gabbro intrusion. From the poorly defined mean direction at site 19 (decl. = 193.8°, incl. = 26.3°, α_{95} = 20.9°) a palaeomagnetic pole position is calculated, which overlaps the 1.3 Ga or 1.6 Ga part of the Fennoscandian APWP. Similar pole positions have been calculated from other dykes in northern and central Sweden (Elming & Mattsson 2001) and it is possible that they form a separate generation of dyke intrusions. The deviating mean direction at site 15 may be a result of deformation and alteration, which is indicated by the anomalous orientation of the foliation plane of the magnetic fabric with respect to the plane of the dyke.

The gabbro carries a characteristic magnetization isolated in high blocking temperatures (≥560°C) and coercivity ranges (≥20 mT). Thermal demagnetizations of the gabbro samples show that the unblocking temperature ranges are generally narrow (Fig. 3C) and the remanence directions in these temperature ranges are consistent, which indicates that the rock is a good recorder of the palaeofield. The slow decay of intensity of magnetization during AF demagnetization (Fig. 3C) suggests that the stable direction of magnetization has its origin in single domain or pseudo-single domain grains (Dunlop & Özdemir 1997). The change of susceptibility with increasing temperatures as well as thermal demagnetizations of the remanent magnetization indicate that magnetite-titanomagnetite is the carrier of remanence.

In low coercivity and unblocking temperature ranges magnetization directions similar to that of the present earth field are often recognized, which suggest a remanence of viscous origin. The characteristic magnetizations form well defined site mean directions (Table 1) and a well fairly defined formation mean (decl. = 329.9°, incl. = 37.7°, α_{95} = 10.3°; Fig. 4A) is determined from the five site means. The pal-

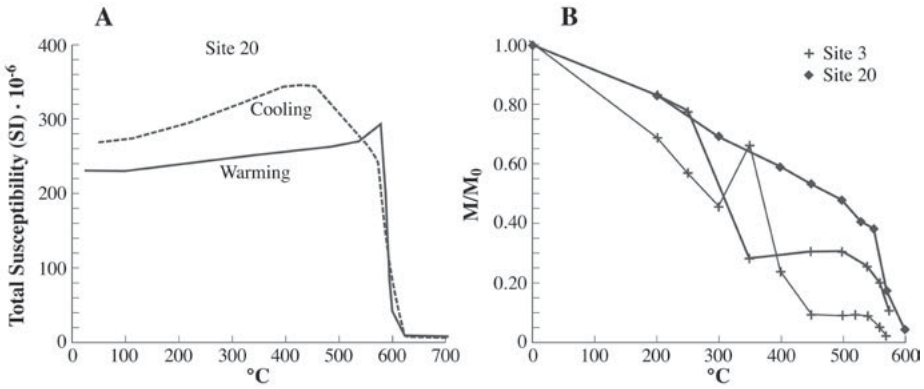


Fig. 5. Exemples of susceptibility **A.** and remanence **B.** behaviour during thermal treatment of samples from the basic dykes. The rapid decrease in susceptibility at c. 580° indicates magnetite to be the dominating magnetic mineral, while the drop and peak (site 3) in remanence intensity at c. 320° suggest that in this site pyrrhotite is a carrier of remenence.

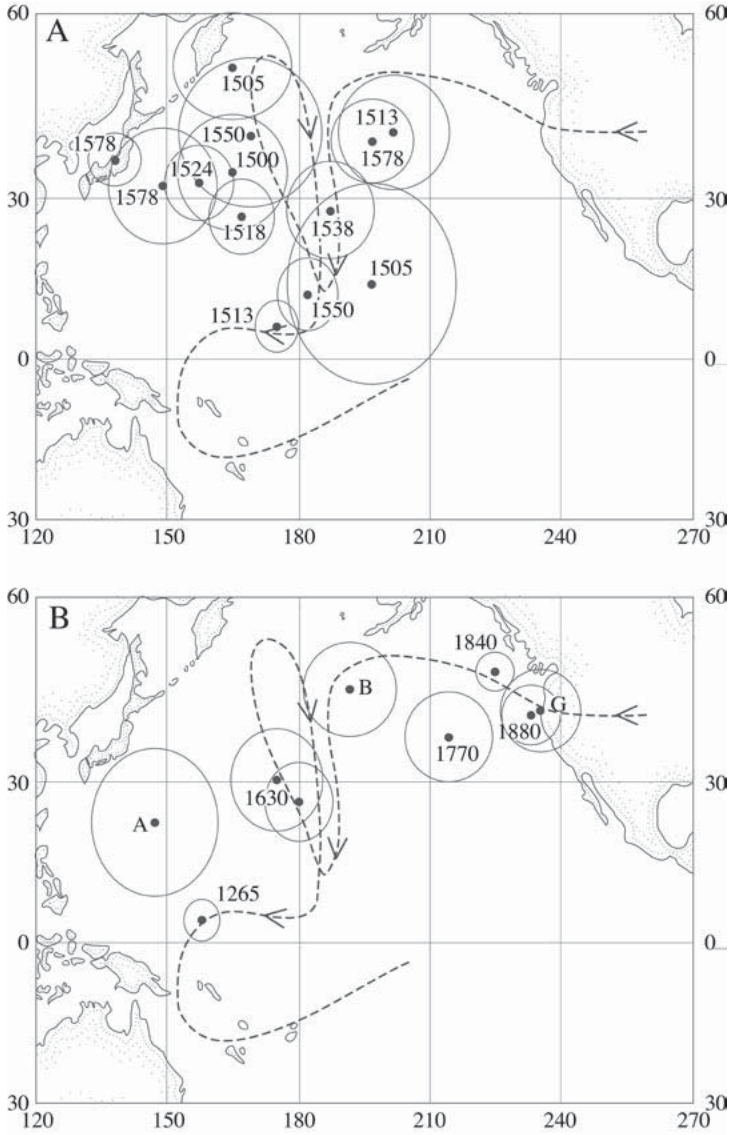


Fig. 6. **A.** Palaeomagnetic pole positions and associated confidence circles (A_{95}) for rapakivi intrusions and related dykes in the Fennoscandian Shield (see Table 2). **B.** Palaeomagnetic poles from the group A and B dykes, and the gabbro (G) of this study and reference poles together with an apparent polar wander path for the Fennoscandian Shield (Elming 1993). The ages are

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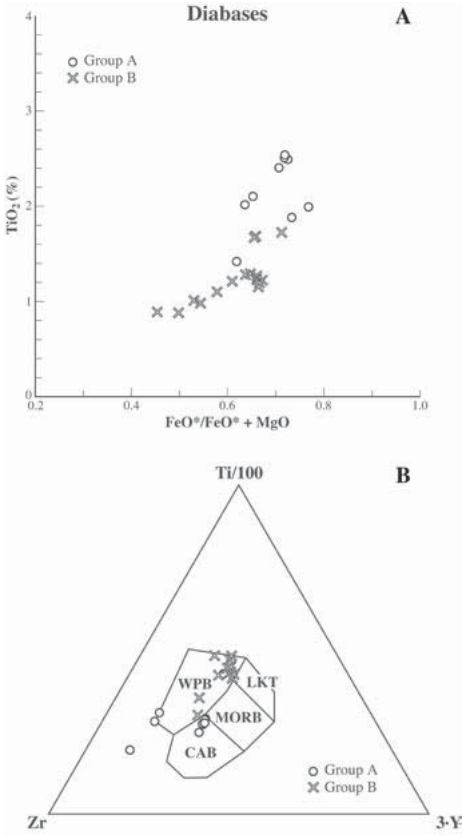


Fig. 7. **A.** FeO*/(FeO*+MgO) vs. TiO_2 for group A and B diabases. FeO* denotes total iron as FeO. **B.** Geochemical classification of group A and B diabases. Diagram after Pearce & Cann (1973).

Table 2. Geochemical data from some of the investigated dykes. Site numbers refer to the palaeomagnetic sites in Fig. 2.

Site	2	3	9	11	12	16	17	20a	22
SiO_2	44.80	44.40	48.10	48.10	47.70	50.20	48.20	51.40	51.80
TiO_2	2.11	2.54	1.73	1.02	1.69	1.11	0.90	2.00	1.24
Al_2O_3	15.40	15.50	14.30	15.60	15.50	15.30	14.70	17.40	14.70
Fe_2O_{3tot}	15.20	16.20	15.20	11.50	15.90	11.70	9.90	10.70	13.00
MnO	0.23	0.24	0.19	0.22	0.22	0.15	0.14	0.13	0.21
MgO	7.26	5.70	5.51	9.16	7.42	7.66	10.70	2.89	6.01
CaO	8.85	8.16	9.40	10.30	8.37	8.38	8.03	4.53	9.26
Na_2O	3.01	3.40	2.93	2.63	2.89	3.19	3.41	4.71	3.16
K_2O	0.56	0.74	0.42	1.33	0.52	0.61	1.56	3.65	0.60
P_2O_5	0.42	0.47	0.22	0.20	0.15	0.12	0.20	0.49	0.12
LOI	2.20	2.20	1.20	2.50	2.30	1.10	2.30	1.80	1.00
SUM	97.80	97.40	98.00	100.00	100.30	98.50	96.60	97.90	100.10
Sr	319	350	485	634	460	421	419	609	499
Ba	327	347	307	552	143	169	429	1020	257
Rb				79	16				24
Nb	11.8	11.9	13.5	5.4	5.4	8.3	16.6	21.9	5.7
Zr	198	258	69	42	68	54	82	247	46
La	13.6	11.9	15.8	11.0	11.0	12.3	16.5	36.7	11.0
Yb	5.8	7.0	3.7	2.1	2.5	2.6	2.8	2.8	2.3
Y	39.5	51.4	19.9	12.0	14.0	12.6	14.6	19.8	13.0
Sc	28.0	25.0	26.5	28	20	20.7	22.6	6.5	25
V	143	121	273	245	160	172	189	93	180
Cr	176	31	97	463	181	276	663	12	70
Ni	110	45	60	138	150	171	187	12	101
Co	47.0	42.1	42.4	52	60	45.7	44.6	23.0	47
Cu	53.8	39.4	127.0	41	64	82.2	37.6	32.5	77
Zn	113	93	131	129	141	101	104	164	113
Pb			12	11	11	12	12	12	11
Mo	5.9	6.0	6.0	5.4	5.4	6.1	5.7	5.9	5.7
W	11.8	11.9	10.0	24.0	26.0	12.3	20.9	11.7	18.0
Be	2.6	2.9	1.2	1.1	1.1	1.2	1.2	2.1	1.1
Sn	28.4	26.4	6.0	5.4	5.4	6.1	5.7	5.9	5.7

Major oxide concentrations are given as Wt%, trace element concentrations as ppm.

aeomagnetc pole position (plat. = 42.5°, plon. = 235.7°, A_{95} = 9.3) calculated from the formation mean falls close to the 1.88 Ga reference pole on the APWP for Fennoscandia (Fig. 6B).

Geochemical analyses of the basic dykes

Chemical analyses have been made on samples from eleven of the twenty sites at which palaeomagnetic dyke samples were collected. One to four samples from each site were analysed.

The chemical data (Table 2) do not show any prominent evidences of alterations and most samples have compositions that occupies the igneous field in the Huges (1973) igneous spectra. However, loss of ignition (LOI) values of up to 3.5% indicate some hydration and low temperature metamorphism of the rocks. The major element composition shows that the basic dykes have a tholeiitic character, which is also supported by the enrichment of Fe and Ti with increasing FeO/(FeO+MgO) ratio (Fig. 7A). The composition of the dykes varies from 44.4% to 53.3% in SiO_2 content and with a Mg-number (Mg#) varying between 70 and 34.

Three rather distinct groups can be distinguished on basis of the main and trace element composition. The first group includes the sites 9, 11, 12, 16 and 22 (palaeomagnetic group B), for which the composition varies from rather primitive to moderately evolved magma (Mg# = 67 to 49), showing a transition from Mg-tholeiites to Fe-tholeiites and with low contents of incompatible elements. In the second group, which includes sites 2 and 3 (palaeomagnetic group A), the composition reflects a moderately to highly evolved magma (Mg# = 55 to 43). The dykes at these sites have a low content of Si and Cr, in combination with high contents of most incompatible elements. The dykes comprising the third group of samples (sites 20 and 21; palaeomagnetic group A) have compositions that are distinctly different from the others as shown by the high contents of incompatible elements in combination with low Mg-numbers (Mg# = 42 to 34) and high contents of Si, Al, and Na.

Two samples from the sites 12 and 17 do not match any of these three compositional groups.

Standard classification diagrams have been used for interpretation of the tectonic setting of the dykes (Figs. 7B and 8). Sam-

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ples from the first group (palaeo-magnetic group B) have a transitional composition between Within Plate Basalt (WPB) and Mid Ocean Ridge Basalt (MORB) and varies from Mg-tholeiites to more evolved Fe-tholeiites with higher contents of incompatible elements. The composition may be controlled by either varying degrees of melting in the source area or differentiation at crustal levels. The dykes at site 2 and 3 (the second group; part of palaeomagnetic group A) have compositions characteristic of WPB. The Nb/Y ratio is low and the trend of moderately to strongly evolved composition may indicate that the dykes were formed by differentiation of rather primitive tholeiitic melts. The dykes of the third group (sites 20 and 21; part of the palaeomagnetic group A) are distinct in their compositional character and plot in the WPB field (Fig. 8). The high Nb/Y ratio (c. 1.0) indicates a slightly alkaline character suggesting an origin by a low degree of mantle melting. However, the high content of Si, Al and incompatible elements may also be a result of differentiation and/or crustal contamination.

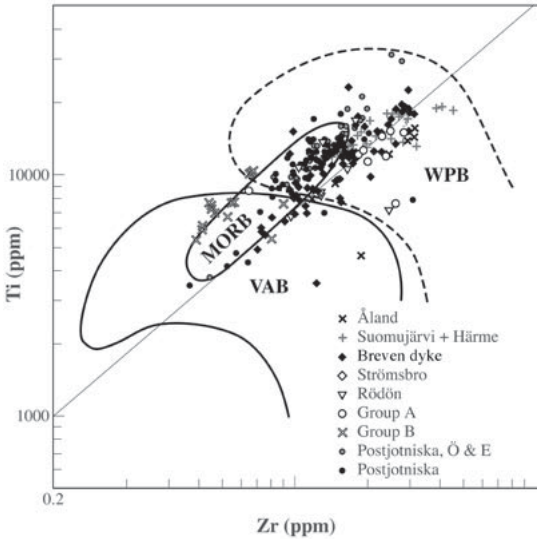


Fig. 8. Geochemical classification of group A and B diabases. Data for postjotnian diabases (Gorbatschev et al. 1979; Lundqvist et al. 1990; Andersson 1997a; Öhlander & Elming unpublished data) and diabases of Rapakivi age (Rämö 1991; Eklund & Lindberg 1992; Lindberg & Eklund 1992; Risku-Norja 1992; Andersson 1997a, 1997b) are included for comparison. Diagram after Pearce (1982).

Discussion

The two main groups of palaeomagnetic directions, group A (sites 2, 3, 20, 21) and B(sites 6, 9, 10, 11, 14, 16, 17, 18, 22), are also characterized by different geochemical compositions, suggesting that the dykes of group A were formed under different geological conditions than those of group B.

Palaeomagnetic studies have previously been performed on the small rapakivi formations south and southeast of the dyke region and palaeomagnetic poles have been calculated from three of them, the Ragunda (Piper 1979), the Mårdsjö (Moakhar & Elming 1998, Moakhar & Elming 2000) and the Nordingrå (Piper 1980a) formations.These formations are located within 100 km from the dyke swarm and they have been dated at 1514–1505 Ma, 1524 and 1578 Ma, respectively (U–Pb; Andersson 1997a; Persson 1999).The chemical compositions of the Ragunda and Mårdsjö formations are somewhat similar to those of site 20 and 21 of group A dykes. A palaeomagnetic pole calculated from one site of the Mårdsjö Formation (Plat = 32.9°, Plon. = 157.3°, A_{95} = 14.8°; Moakhar & Elming 2000; Table 3) is also similar to the poorly defined mean pole of group A (Plat. = 21.6°, Plon. = 147.1°, A_{95} = 14.8°; Fig. 6A) but different from the pole of the Ragunda Formation (Plat. = 51.6°, Plon. = 166.6°, A_{95} = 7.1°; Piper 1979) and the very poorly defined pole of the related dykes (Plat. = 64.3°, Plon. = 154.2°, A_{95} = 24.7°). This difference may possibly be related to the 10 to 20 Ma of age difference between the formations. However, poles calculated for the significantly older 1578 Ga Nordingrå complex (granite, Plat. = 32.4°, Plon. = 149°; gabbro, Plat. = 36.9°, Plon. = 138.1°; Piper 1980) are not far from the mean pole of the group A dykes.

The mean pole for the group B dykes (Plat. = 45.2°, Plon. = 191.5°, A_{95} = 7.5°) falls close to the mean pole of the Nordingrå basic dykes (Plat. = 40°, Plon. = 197°; Table 3) and to the pole of the dykes related to the 1497–1513 Ma old Rödö rapakivi complex (Plat. = 41.6°, Plon. = 201.7°; Moakhar & Elming 2000). The pole of the 1523–1553 Ma old Föglö-Sottunga dykes (Plat. = 27.9°, Plon. = 187.5°; Pesonen & Neuvonen 1981) is not that different from the pole of group B either.

The poles calculated from 1578 to 1505 Ma old rapakivis in Sweden (Table 3; Fig. 6A) are quite scattered and fall between ca 150 to 200° in longitude and c. 30 to 64° in latitude. This may to some extent be a result of poor palaeomagnetic data as sometimes expressed by the big A_{95}, but may also indicate a complex apparent polar wander during this time.

Comparing the group A and B poles with the reference poles for 1.77, 1.63 and 1.27 Ga (Table 4; Fig. 6B) the pole B falls in between the 1.77 and the 1.63 Ga reference poles, while the group A pole falls between the 1.63 and the 1.25 Ga poles. This may suggest that the group A dykes are younger that the group B dykes. However, keeping in mind the big scatter in pole positions for rapakivi related rocks of similar ages, this dating based on the comparison of poles is not very reliable.

The group A dykes have a geochemical signature typical for intraplate settings and the slightly alkaline character of the magmas suggests rather low degree of mantle melting. The bimodal Rapakivi magmatism is generally attributed to mantle upwelling during crustal extension and thinning that generated mafic magmas of mantle origin and felsic magmas by melting of underplated lower crust (e.g. Haapala & Rämö 1992; Korja & Heikinen 1995; Andersson 1997a; Persson 1999; Puura & Flodén 2000). The geochemical signature of the mafic dykes in this study indicates magma modification by fractionation and

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contamination at crustal levels. This suggests a tectonic regime of mainly weak extension prohibiting rapid magma ascent through the crust in accordance with the importance of mafic underplating in the rapakivi mag-matism (Korja & Heikinen 1995; Puura & Flodén 2000). The group B dykes are more primitive in composition compared to the rapakivi dykes and may have formed by higher degree of mantle melting due to extensive heating and/or decompression of the upper mantle. The limited signs of crustal contamination and their mostly rather unfractionated character suggest rapid magma ascent to upper crustal levels during strong extensional stress.

A temporal and spatial relation between the accretional southwestward growth of the Baltic Shield and the formation of the rapakivi suites have been proposed by Åhäll et al. (2000). They suggested that the westward stepwise movement of the subduction beneath Baltica resulted in changing intracratonic lithospheric stress, which may have guided the timing and location of the rapakivi magmatism. The melt, that may have been generated in the lower lithosphere by decompressional melting, ascent through the crust possibly guided by relict fractures. The NE to ENE trend of the basic dykes is more or less perpendicular to the growth zones and parallell to an expected palaeocompressional stress field. Such a stress field, related to the beginning of the Gothian orogeny, was also suggested to be the origin of a major deformation zone in central Sweden (Mattsson & Elming 2001). Dykes have been used as palaeostress indicators in areas of continental collision zones (Féraud et al. 1987), and they were found to be intruded normal to the collision zone and parallel to the vertical plane containing the maximum principal stress field. A trend similar to that of the dykes in this study is also noticed for the Åland rapakivi dykes (Fig. 1), while rapakivi dykes of other generations are generally perpendicular to that trend.

The difference in geochemical character between group A and B dykes (Figs. 7 and 8) supports the interpretation of the palaeomagnetic data that suggest a different time of intrusion of these dykes. However, with respect to the chemical data the dykes in group A are not homogeneous. The composition of the dykes at the sites 2 and 3 (Fig. 8) shows similarities with the composition of rapakivi related mafic dykes from southern Finland (Åland and Suomenniemi dykes; Rämö 1991; Eklund & Lindberg 1992; Lindberg & Eklund 1992) and central Sweden (Breven dyke; Risku-Norja 1992; Ströms-bro dykes; Andersson 1997b), while the dykes at the sites 20 and 21 have compositions similar to the rapakivi dyke at Rödön, central Sweden (Andersson 1997a).

Postjotnian, c. 1.26 Ga, sills and dykes are widespread in central and northern Sweden (Gorbatschev et al. 1979; Elming & Mattsson 2001). The composition of these basic intrusions are clearly different from the rapakivi dykes and in most compositional diagrams they fall between the compositions of group A and groups B.

The latitudinal position and orientation as calculated from the group B pole position suggest that Fennoscandia and the Baltic Shield was located in a shallow northern latitude with an orientation not significantly different from that at present. The palaeoposition calculated from the group A pole indicates that Baltica was located very close to the equator at the time of the intrusion of the dykes. Baltica had then an orientation corresponding to counterclockwise rotation of c. 45°, with respect to the present orientation (Fig. 9A).

Based on palaeomagnetic data from Fennoscandia tectonical reconstructions have been made for the Palaeo- and Mesoproterozoic (e.g. Pesonen et al. 1989; Elming et al. 1993). From comparison with the drift of other continents Pesonen et al. (2001, 2003) has suggested that Fennoscandia and Laurentia were joined from c. 1.88 Ga to 1.25 Ga. The two continents then drifted from shallow northern latitudes towards shallow southern latitudes and meanwhile they rotated counterclockwise. Assuming that the group B pole is older than that of group A the palaeopositions calculated for Fenno-scandia from these poles falls into this drift pattern (Fig. 9A). In this time period the Ukrainian Shield accreted to Fenno-scan-

Table 3. Palaeomagnetic pole positions from rapakivi formations and associated dykes in the Fennoscandian Shield.

Code	Rock	Glat./Glon.	N/n	Plat./Plon.	P	A_{95}	Age(Ma)	Ref
RD	Rödö Dykes	62.4/17.6	6*/53	41.6/201.7	r	9.5	1513	1,2
RG	Rödö granite	62.4/17.6	6*/20	6.2/175.0	n	4.8	1513	1,2
Må	Mårdsjö Gabbro	63.8/15.7	1/7*	32.9/157.3	n	6.5	1524	1,2
RC	Ragunda Complex	63.3/16.1	15*/87	51.6/166.6	c	7.1	1505–1514	3
RD	Ragunda Basic dykes(1)	63.3/16.1	12*/64	14.6/194.4	c	11.1	1505–1514	3
RB	Ragunda Basic dykes(2)	63.3/16.1	4*/24	64.3/154.2	c	24.7	1505–1514	3
NG	Nordingrå Granite	62.9/18.6	28*/28	32.4/149.0	c	10.5	1578	4
NO	Nordingrå Gabbro	62.9/18.6	19*/108	36.9/138.1	c	4.7	1578	4
ND	Nordingrå Basic dykes	62.9/18.4	7*/7	40.0/197.0	n	7.1	1578?	4
GG	Gävle Granite	60.7/17.2	12*/12	34.7/165.0	n	10.0	1500	4
Hä	Hälllefors Basic dyke	59.1/16.7	1/13*	27.0/167.0	n	6.7	1518	5
FD	Föglö-Sott. Dykes	60.0/20.6	6*/25	27.9/187.5	n	8.8	1523–1553	6
KD	Kumlinge-Brändö Dykes	60.3/20.8	7*/25	12.2/182.0	r	6.7	1550	6
ÅI	Mean Åva Intrusion	60.6/21.1	15*/15	41.0/169.0	n	12.4	1550	7

N/n – number of sites/number of samples; P – polarity; Age – U–Pb ages of the formations; Ref – references to the palaeomagnetic studies: (1) Moakhar & Elming 1998, (2) Moakhar & Elming 2000, (3) Piper 1979, (4) Piper 1980a, (5) Piper 1980b, (6) Pesonen & Neuvonen 1981, (7) Neuvonen 1970. For other explanations see Table 1.

Table 4. Reference poles for the Baltic Shield. For explanation see Table 1.

Rock	Plat.	Plon.	A_{95}	Age (Ma)	Reference
Vittangi gabbro	43.0	228.0	5.0	1886	Elming 1985
Kiuruvesi gabbro	41.0	231.0	4.0	1886	Neuvonen et al. 1981
Pohjanmaa gabbro	38.0	239.0	11.0	1879	Pesonen & Stigzelius 1972
Jalokoski gabbro	43.0	234.0	7.0	1871	Mertanen & Pesonen 1992
Mean	**41.0**	**233.0**	**5.0**	**1880**	
Haukivesi dykes	48.0	225.0	3.0	**1840**	Neuvonen et al. 1981
Ropruchey-Shosksha Fm.	38.3	214.5	8.0	**1770**	Khramov et al. 2000
SE quartz porphyry dykes	30.0	175.0	9.0	1638–1617	Neuvonen 1986
Sipoo quartz porphyry dykes	26.0	180.0	7.0	1630	Mertanen & Pesonen 1994
Mean	**29.0**	**177.0**		**1630**	
Vaasa dolerite	7.0	164.0	4.0	1268	Neuvonen 1966
Satakunta dolerite	2.0	158.0	4.0	1264	Neuvonen 1965
Märket dolerite	–6.0	146.0	9.0	1265	Neuvonen & Grundström 1969
Mean	**4.0**	**158.0**	**4.0**	**1265**	

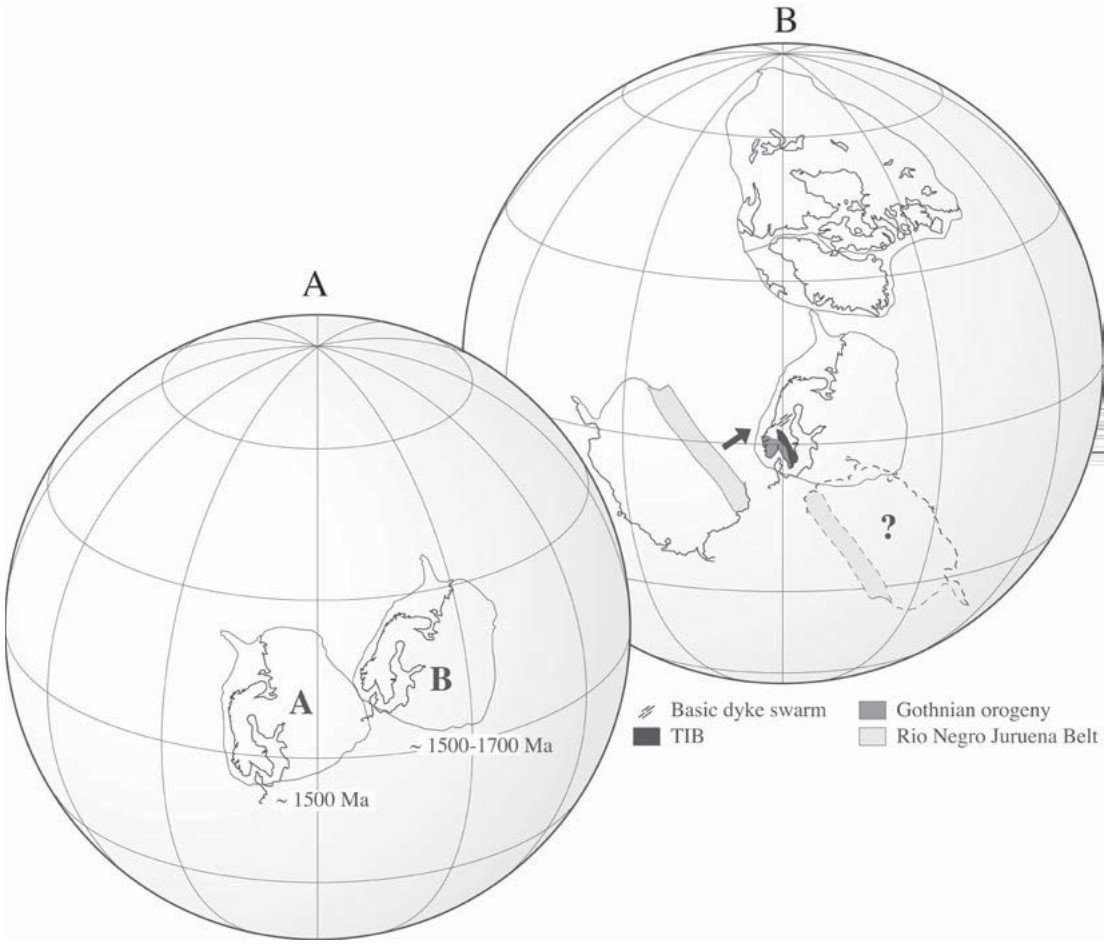


Fig. 9. **A.** The reconstructed positions of Baltica as calculated from the group A and B mean poles. **B.** The tectonic reconstruction of Baltica with reference to Laurentia and Amazonia at 1650 Ma (modified after Elming et al. 2001, and Pesonen et al. 2001). The arrow marks the estimated direction of compression, with reference to the soutwestward crustal growth of Fennoscandia. Note that the TIB and the Gothian Belt in Baltica in this reconstruction are parallell and facing the coeval Rio Negro Juruena Belt in Amazonia.

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dia and the Baltic Shield was formed in a supercontinental assembly with Laurentia (Fig. 9B), a supercontinet named Hudsonland (Williams et al. 1991; Pesonen et al. 2003). The dykes most probably intruded during and slightly after the time of the Gothian orogeny (1.75–1.55 Ga). It has been suggested that the intrusions of the Transscandinavian Igneous Belt (1.81–1.66 Ga, Åhäll & Larsson 2000) and the Gothian orogeny were the result of an island arc-accretion followed by continental collision with the Amazonian Shield docked to the soutwestern Fennoscandia/Laurentia (Fig. 9B; Elming et al. 2001). However, this scenario is difficult to argue for on geological grounds and another model has been presented in which Amazonia is located south of Fennoscandia, aligning coeval belts in Laurentia, Baltica and Amazonia (e.g. Bettencourt et al. 1996; Åhäll & Larsson 2000; Geraldes et al. 2001). This model is however not justified by palaeomagnetic data, but can not be rejected as the palaeomagnetic data and age data are not very well constrained.

Conclusion

There are two generations of NE–SW to E–W trending dykes in the Åsele region in northern Sweden. This is concluded from the different palaeomagnetic signatures as well as compositional differences. The dykes of group A have compositions that are similar to rapakivi related dykes, while those of group B are different from most rapakivi dykes in the Fennoscandian Shield due to their more primitive character and low contents of incompatible elements. The calculated pole positions may suggest that the group B dykes are older than the group A dykes and both systems of dykes are probably intruded within the time span 1.77 Ga to 1.5 Ga. The NE to ENE trend of the dykes is more or less parallel to the palaeo-compressional stress field expected from the collisional tectonics in southwestern Sweden. The intrusion of the rapakivi formations has been suggested to be related with the Gothian orogeny and the intrusion of the dykes may thus be guided by the stress field generated during the stepwise westward crustal growth of Baltica. The dykes were intruded when Baltic formed a part of a supercontinent, Hudsonland, that included Laurentia and possibly also Amazonia.

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